CHAPTER 6

ACCELERATED OBSERVERS

The objective world simply is; *it does not* happen. *Only to the gaze of my consciousness, crawling upward along the life line* [*world line*] *of my body, does a section of this world come to life as a fleeting image in space which continuously changes in time.*

HERMAN WEYL (1949, p. 116)

§6.1. ACCELERATED OBSERVERS CAN BE ANALYZED USING SPECIAL RELATIVITY

It helps in analyzing gravitation to consider a situation where gravity is mocked up by acceleration. Focus attention on a region so far from any attracting matter, and so free of disturbance, that (to some proposed degree of precision) spacetime there can be considered to be flat and to have Lorentz geometry. Let the observer acquire the feeling that he is subject to gravity, either because of jet rockets strapped to his legs or because he is in a rocket-driven spaceship. How does physics look to him?

Dare one answer this question? At this early stage in the book, is one not too ignorant of gravitation physics to predict what physical effects will be measured by an observer who thinks he is in a gravitational field, although he is really in an accelerated spaceship? Quite the contrary; special relativity was developed precisely to predict the physics of accelerated objects—e.g., the radiation from an accelerated charge. Even the fantastic accelerations

Accelerated motion and accelerated observers can be analyzed using special relativity

$$a_{\text{nuclear}} \sim v^2/r \sim 10^{31}\ \text{cm/sec}^2 \sim 10^{28}\ \text{“earth gravities”}$$

suffered by a neutron bound in a nucleus, and the even greater accelerations met in high-energy particle-scattering events, are routinely and accurately treated within

Box 6.1 GENERAL RELATIVITY IS BUILT ON SPECIAL RELATIVITY

A tourist in a powered interplanetary rocket feels "gravity." Can a physicist by local effects convince him that this "gravity" is bogus? Never, says Einstein's principle of the local equivalence of gravity and accelerations. But then the physicist will make no errors if he deludes himself into treating true gravity as a local illusion caused by acceleration. Under this delusion, he barges ahead and solves gravitational problems by using special relativity: if he is clever enough to divide every problem into a network of local questions, each solvable under such a delusion, then he can work out all influences of any gravitational field. Only three basic principles are invoked: special-relativity physics, the equivalence principle, and the local nature of physics. They are simple and clear. To apply them, however, imposes a double task: (1) take spacetime apart into locally flat pieces (where the principles are valid), and (2) put these pieces together again into a comprehensible picture. To undertake this dissection and reconstitution, to see curved dynamic spacetime inescapably take form, and to see the consequences for physics: that is general relativity.

the framework of special relativity. The theoretician who confidently applies special relativity to antiproton annihilations and strange-particle resonances is not about to be frightened off by the mere illusions of a rocket passenger who gullibly believed the travel brochures advertising "earth gravity all the way." When spacetime is flat, move however one will, special relativity can handle the job. (It can handle bigger jobs too; see Box 6.1.) The essential features of *how* special relativity handles the job are summarized in Box 6.2 for the benefit of the Track-1 reader, who can skip the rest of the chapter, and also for the benefit of the Track-2 reader, who will find it useful background for the rest of the chapter.

Box 6.2 ACCELERATED OBSERVERS IN BRIEF

An accelerated observer can carry clocks and measuring rods with him, and can use them to set up a reference frame (coordinate system) in his neighborhood.

His clocks, if carefully chosen so their structures are affected negligibly by acceleration (e.g., atomic clocks), will tick at the same rate as unaccelerated clocks moving momentarily along with him:

$$\Delta\tau \equiv \begin{pmatrix} \text{time interval ticked off} \\ \text{by observer's clocks as he} \\ \text{moves a vector displacement} \\ \boldsymbol{\xi} \text{ along his world line} \end{pmatrix} = [-\boldsymbol{g}(\boldsymbol{\xi}, \boldsymbol{\xi})]^{1/2}.$$

And his rods, if chosen to be sufficiently rigid, will measure the same lengths as momentarily comoving, unaccelerated rods do. (For further discussion, see §16.4, and Boxes 16.2 to 16.4.)

Let the observer's coordinate system be a Cartesian latticework of rods and clocks, with the origin of the lattice always on his world line. He must keep his latticework small:

$$l \equiv \begin{pmatrix}\text{spatial dimensions}\\ \text{of lattice}\end{pmatrix} \ll \begin{pmatrix}\text{the acceleration measured}\\ \text{by accelerometers he carries}\end{pmatrix}^{-1} \equiv \frac{1}{g}.$$

At distances l away from his world line, strange things of dimensionless magnitude gl happen to his lattice—e.g., the acceleration measured by accelerometers differs from g by a fractional amount $\sim gl$ (exercise 6.7); also, clocks initially synchronized with the clock on his world line get out of step (tick at different rates) by a fractional amount $\sim gl$ (exercise 6.6). (Note that an acceleration of one "earth gravity" corresponds to

$$g^{-1} \sim 10^{-3}\ \text{sec}^2/\text{cm} \sim 10^{18}\ \text{cm} \sim 1\ \text{light-year},$$

so the restriction $l \ll 1/g$ is normally not severe.)

To deduce the results of experiments and observations performed by an accelerated observer, one can analyze them in coordinate-independent, geometric terms, and then project the results onto the basis vectors of his accelerated frame. Alternatively, one can analyze the experiments and observations in a Lorentz frame, and then transform to the accelerated frame.

As deduced in this manner, the results of experiments performed locally (at $l \ll 1/g$) by an accelerated observer differ from the results of the same experiments performed in a Lorentz frame in only three ways:

(1) There are complicated fractional differences of order $gl \ll 1$ mentioned above, that can be made negligible by making the accelerated frame small enough.

(2) There are Coriolis forces of precisely the same type as are encountered in Newtonian theory (exercise 6.8). These the observer can get rid of by carefully preventing his latticework from rotating—e.g., by tying it to gyroscopes that he accelerates with himself by means of forces applied to their centers of mass (no torque!). Such a nonrotating latticework has "Fermi-Walker transported" basis vectors (§6.5),

$$\frac{d\boldsymbol{e}_{\alpha'}}{d\tau} = \boldsymbol{u}(\boldsymbol{a}\cdot\boldsymbol{e}_{\alpha'}) - \boldsymbol{a}(\boldsymbol{u}\cdot\boldsymbol{e}_{\alpha'}), \tag{1}$$

where $\boldsymbol{u}$ = 4-velocity, and $\boldsymbol{a} = d\boldsymbol{u}/d\tau$ = 4-acceleration.

(3) There are inertial forces of precisely the same type as are encountered in Newtonian theory (exercise 6.8). These are due to the observer's acceleration, and he cannot get rid of them except by stopping his accelerating.

§6.2. HYPERBOLIC MOTION

The rest of this chapter is Track 2.
It depends on no preceding Track-2 material.
It is needed as preparation for
(1) the mathematical analysis of gyroscopes in curved spacetime (exercise 19.2, §40.7), and
(2) the mathematical theory of the proper reference frame of an accelerated observer (§13.6).
It will be helpful in many applications of gravitation theory (Chapters 18–40).

Study a rocket passenger who feels "gravity" because he is being accelerated in flat spacetime. Begin by describing his motion relative to an inertial reference frame. His 4-velocity satisfies the condition $\boldsymbol{u}^2 = -1$. To say that it is fixed in magnitude is to say that the 4-acceleration,

$$\boldsymbol{a} = d\boldsymbol{u}/d\tau, \tag{6.1}$$

is orthogonal to the 4-velocity:

$$0 = (d/d\tau)(-1/2) = (d/d\tau)\left(\frac{1}{2}\boldsymbol{u}\cdot\boldsymbol{u}\right) = \boldsymbol{a}\cdot\boldsymbol{u}. \tag{6.2}$$

This equation implies that $a^0 = 0$ in the rest frame of the passenger (that Lorentz frame, where, at the instant in question, $\boldsymbol{u} = \boldsymbol{e}_0$); in this frame the space components of a^μ reduce to the ordinary definition of acceleration, $a^i = d^2x^i/dt^2$. From the components $a^\mu = (0; a^i)$ in the rest frame, then, one sees that the magnitude of the acceleration in the rest frame can be computed as the simple invariant

$$\boldsymbol{a}^2 = a^\mu a_\mu = (d^2\boldsymbol{x}/dt^2)^2{}_{\text{as measured in rest frame}}.$$

Consider, for simplicity, an observer who feels always a constant acceleration g. Take the acceleration to be in the x^1 direction of some inertial frame, and take $x^2 = x^3 = 0$. The equations for the motion of the observer in that inertial frame become

$$\frac{dt}{d\tau} = u^0, \qquad \frac{dx}{d\tau} = u^1; \qquad \frac{du^0}{d\tau} = a^0, \qquad \frac{du^1}{d\tau} = a^1. \tag{6.3}$$

Write out the three algebraic equations

$$u^\mu u_\mu = -1,$$
$$u^\mu a_\mu = -u^0a^0 + u^1a^1 = 0,$$

and

$$a^\mu a_\mu = g^2.$$

Solve for the acceleration, finding

$$a^0 = \frac{du^0}{d\tau} = gu^1, \qquad a^1 = \frac{du^1}{d\tau} = gu^0. \tag{6.4}$$

These linear differential equations can be solved immediately. The solution, with a suitable choice of the origin, reads

$$t = g^{-1}\sinh g\tau, \qquad x = g^{-1}\cosh g\tau. \tag{6.5}$$

Uniformly accelerated observer moves on hyperbola in spacetime diagram

Note that $x^2 - t^2 = g^{-2}$. The world line is a hyperbola in a spacetime diagram ("hyperbolic motion"; Figure 6.1). Several interesting aspects of this motion are

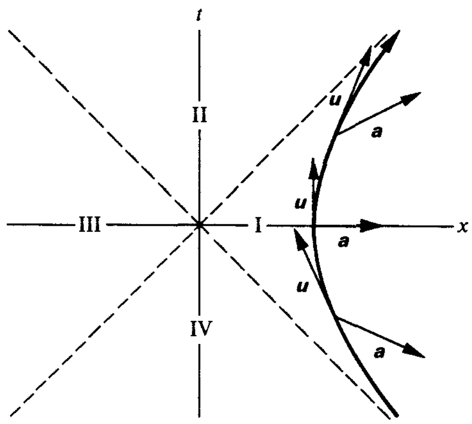

Figure 6.1.
Hyperbolic motion. World line of an object that (or an observer who) experiences always a fixed acceleration g with respect to an inertial frame that is instantaneously comoving (different inertial frames at different instants!). The 4-acceleration $\boldsymbol{a}$ is everywhere orthogonal (Lorentz geometry!) to the 4-velocity $\boldsymbol{u}$.

treated in the exercises. Let the magnitude of the constant acceleration g be the acceleration of gravity, $g = 980\ \text{cm/sec}^2$ experienced on earth: $g \simeq (10^3\ \text{cm/sec}^2)/(3 \times 10^{10}\ \text{cm/sec})^2 = (3 \times 10^7\ \text{sec} \cdot 3 \times 10^{10}\ \text{cm/sec})^{-1} = (1\ \text{light-year})^{-1}$. Thus the observer will attain relativistic velocities after maintaining this acceleration for something like one year of his own proper time. He can outrun a photon if he has a head start on it of one light-year or more.

EXERCISES

Exercise 6.1. A TRIP TO THE GALACTIC NUCLEUS

Compute the proper time required for the occupants of a rocket ship to travel the ~30,000 light-years from the Earth to the center of the Galaxy. Assume that they maintain an acceleration of one "earth gravity" ($10^3\ \text{cm/sec}^2$) for half the trip, and then decelerate at one earth gravity for the remaining half.

Exercise 6.2. ROCKET PAYLOAD

What fraction of the initial mass of the rocket can be payload for the journey considered in exercise 6.1? Assume an ideal rocket that converts rest mass into radiation and ejects all the radiation out the back of the rocket with 100 per cent efficiency and perfect collimation.

Exercise 6.3. TWIN PARADOX

(a) Show that, of all timelike world lines connecting two events $\mathscr{A}$ and $\mathscr{B}$, the one with the *longest* lapse of proper time is the unaccelerated one. (*Hint:* perform the calculation in the inertial frame of the unaccelerated world line.)

(b) One twin chooses to move from $\mathscr{A}$ to $\mathscr{B}$ along the unaccelerated world line. Show that the other twin, by an appropriate choice of accelerations, can get from $\mathscr{A}$ to $\mathscr{B}$ in arbitrarily small proper time.

(c) If the second twin prefers to ride in comfort, with the acceleration he feels never exceeding one earth gravity, g, what is the shortest proper time-lapse he can achieve between $\mathscr{A}$ and $\mathscr{B}$? Express the answer in terms of g and the proper time-lapse $\Delta\tau$ measured by the unaccelerated twin.

(d) Evaluate the answer numerically for several interesting trips.

Exercise 6.4. RADAR SPEED INDICATOR

A radar set measures velocity by emitting a signal at a standard frequency and comparing it with the frequency of the signal reflected back by another object. This redshift measurement is then converted, using the standard special-relativistic formula, into the corresponding velocity, and the radar reads out this velocity. How useful is this radar set as a velocity-measuring instrument for a uniformly accelerated observer?

(a) Consider this problem first for the special case where the object and the radar set are at rest with respect to each other at the instant the radar pulse is reflected. Compute the redshift $1 + z = \omega_e/\omega_0$ that the radar set measures in this case, and the resulting (incorrect) velocity it infers. Simplify by making use of the symmetries of the situation.

(b) Now consider the situation where the object has a non-zero velocity in the momentary rest frame of the observer at the instant it reflects the radar pulse. Compute the ratio of the actual relative velocity to the velocity read out by the radar set.

Exercise 6.5. RADAR DISTANCE INDICATOR

Use radar as a distance-measuring device. The radar set measures its proper time τ between the instant at which it emits a pulse and the later instant when it receives the reflected pulse. It then performs the simple computation $L_0 = \tau/2$ and supplies as output the "distance" L_0. How accurate is the output reading of the radar set for measuring the actual distance L to the object, when used by a uniformly accelerated observer? (L is defined as the distance in the momentary rest frame of the observer at the instant the pulse is reflected, which is at the observer's proper time halfway between emitting and receiving the pulse.) Give a correct formula relating $L_0 \equiv \tau/2$ to the actual distance L. Show that the reading L_0 becomes infinite as L approaches g^{-1}, where g is the observer's acceleration, as measured by an accelerometer he carries.

§6.3. CONSTRAINTS ON SIZE OF AN ACCELERATED FRAME

Difficulties in constructing "the coordinate system of an accelerated observer":

It is very easy to put together the words "the coordinate system of an accelerated observer," but it is much harder to find a concept these words might refer to. The most useful first remark one can make about these words is that, if taken seriously, they are self-contradictory. The definite article "the" in this phrase suggests that one is thinking of some unique coordinate system naturally associated with some specified accelerated observer, such as one whose world line is given in equation (6.5). If the coordinate system is indeed natural, one would expect that the coordinates of any event could be determined by a sufficiently ingenious observer by sending and receiving light signals. But from Figure 6.1 it is clear that the events composing one quarter of all spacetime (Zone III) can neither send light signals to, nor receive light signals from, the specified observer. Another half of spacetime suffers lesser disabilities in this respect: Zone II cannot send to the observer, Zone IV cannot receive from him. It is hard to see how the observer could define in any natural way a coordinate system covering events with which he has no causal relationship, which he cannot see, and from which he cannot be seen!

Breakdown in communication between observer and events at distance $l >$ (acceleration)$^{-1}$

Difficulties also occur when one considers an observer who begins at rest in one frame, is accelerated for a time, and maintains thereafter a constant velocity, at rest in some other inertial coordinate system. Do his motions define in any natural way

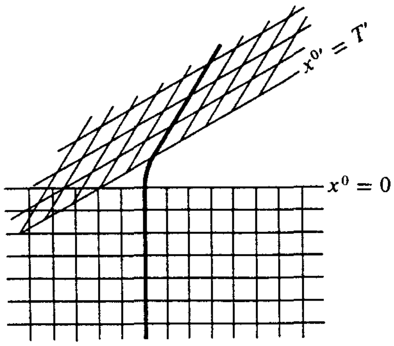

Figure 6.2.
World line of an observer who has undergone a brief period of acceleration. In each phase of motion at constant velocity, an inertial coordinate system can be set up. However, there is no way to reconcile these discordant coordinates in the region of overlap (beginning at distance g^{-1} to the left of the region of acceleration).

a coordinate system? Then this coordinate system (1) should be the inertial frame x^μ in which he was at rest for times x^0 less than 0, and (2) should be the other inertial frame $x^{\mu'}$ for times $x^{0'} > T'$ in which he was at rest in that other frame. Evidently some further thinking would be required to decide how to define the coordinates in the regions not determined by these two conditions (Figure 6.2). More serious, however, is the fact that these two conditions are inconsistent for a region of spacetime that satisfies simultaneously $x^0 < 0$ and $x^{0'} > T'$. In both examples of accelerated motion (Figures 6.1 and 6.2), the serious difficulties about defining a coordinate system begin only at a finite distance g^{-1} from the world line of the accelerated observer. The problem evidently has no solution for distances from the world line greater than g^{-1}. It does possess a natural solution in the immediate vicinity of the observer. This solution goes under the name of "Fermi-Walker transported orthonormal tetrad." The essential idea lends itself to simple illustration for hyperbolic motion, as follows.

Natural coordinates inconsistent at distance $l >$ (acceleration)$^{-1}$

§6.4. THE TETRAD CARRIED BY A UNIFORMLY ACCELERATED OBSERVER

An infinitesimal version of a coordinate system is supplied by a "tetrad," or "moving frame" (Cartan's "repère mobile"), or set of basis vectors $\boldsymbol{e}_{0'}$, $\boldsymbol{e}_{1'}$, $\boldsymbol{e}_{2'}$, $\boldsymbol{e}_{3'}$ (subscript tells which vector, not which component of one vector!) Let the time axis be the time axis of a comoving inertial frame in which the observer is momentarily at rest. Thus the zeroth basis vector is identical with his 4-velocity: $\boldsymbol{e}_{0'} = \boldsymbol{u}$. The space axes $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$ are not affected by Lorentz transformations in the 1-direction. Therefore take $\boldsymbol{e}_{2'}$ and $\boldsymbol{e}_{3'}$ to be the unit basis vectors of the all-encompassing Lorentz frame relative to which the hyperbolic motion of the observer has already been described in equations (6.5): $\boldsymbol{e}_{2'} = \boldsymbol{e}_2$; $\boldsymbol{e}_{3'} = \boldsymbol{e}_3$. The remaining basis vector, $\boldsymbol{e}_{1'}$, orthogonal to the other three, is parallel to the acceleration vector, $\boldsymbol{e}_{1'} = g^{-1}\boldsymbol{a}$ [see equation (6.4)]. There is a more satisfactory way to characterize this moving frame: the time axis $\boldsymbol{e}_{0'}$ is the observer's 4-velocity, so he is always at rest in this frame; and the

Orthonormal tetrad of basis vectors carried by uniformly accelerated observer

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other three vectors $\boldsymbol{e}_{i'}$ are chosen in such a way as to be (1) orthogonal and (2) nonrotating. These basis vectors are:

$$
\begin{aligned}
(e_{0'})^{\mu} &= (\cosh g\tau;\ \sinh g\tau,\ 0,\ 0);\\
(e_{1'})^{\mu} &= (\sinh g\tau;\ \cosh g\tau,\ 0,\ 0);\\
(e_{2'})^{\mu} &= (0;\ 0,\ 1,\ 0);\\
(e_{3'})^{\mu} &= (0;\ 0,\ 0,\ 1).
\end{aligned} \tag{6.6}
$$

There is a simple prescription to obtain these four basis vectors. Take the four basis vectors $\boldsymbol{e}_0$, $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$ of the original global Lorentz reference frame, and apply to them a simple boost in the 1-direction, of such a magnitude that $\boldsymbol{e}_{0'}$ comes into coincidence with the 4-velocity of the observer. The fact that these vectors are all orthogonal to each other and of unit magnitude is formally stated by the equation

$$\boldsymbol{e}_{\mu'} \cdot \boldsymbol{e}_{\nu'} = \eta_{\mu'\nu'}. \tag{6.7}$$

§6.5. THE TETRAD FERMI-WALKER TRANSPORTED BY AN OBSERVER WITH ARBITRARY ACCELERATION

Orthonormal tetrad of arbitrarily accelerated observer: should be "nonrotating"

Turn now from an observer, or an object, in hyperbolic motion to one whose acceleration, always finite, varies arbitrarily with time. Here also we impose three criteria on the moving, infinitesimal reference frame, or tetrad: (1) the basis vectors $\boldsymbol{e}_{\mu'}$ of the tetrad must remain orthonormal [equation (6.7)]; (2) the basis vectors must form a rest frame for the observer at each instant ($\boldsymbol{e}_{0'} = \boldsymbol{u}$); and (3) the tetrad should be "nonrotating."

This last criterion requires discussion. The basis vectors of the tetrad at any proper time τ must be related to the basis vectors $\boldsymbol{e}_0$, $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$ of some given inertial frame by a Lorentz transformation $\boldsymbol{e}_{\mu'}(\tau) = \Lambda^{\nu}{}_{\mu'}(\tau)\boldsymbol{e}_{\nu}$. Therefore the basis vectors at two successive instants must also be related to each other by a Lorentz transformation. But a Lorentz transformation can be thought of as a "rotation" in spacetime. The 4-velocity $\boldsymbol{u}$, always of unit magnitude, changes in direction. The very concept of acceleration therefore implies "rotation" of velocity 4-vector. How then is the requirement of "no rotation" to be interpreted? Demand that the tetrad $\boldsymbol{e}_{\mu'}(\tau)$ change from instant to instant by precisely that amount implied by the rate of change of $\boldsymbol{u} = \boldsymbol{e}_{0'}$, and by no additional arbitrary rotation. In other words, (1) accept the inevitable pseudorotation in the timelike plane defined by the velocity 4-vector and the acceleration, but (2) rule out any ordinary rotation of the three space vectors.

"Nonrotating" means rotation only in timelike plane of 4-velocity and 4-acceleration

Mathematics of rotation in 3-space

Nonrelativistic physics describes the rotation of a vector (components v_i) by an instantaneous angular velocity vector (components ω_i). This angular velocity appears in the formula for the rate of change of $\boldsymbol{v}$,

$$(dv_i/dt) = (\boldsymbol{\omega} \times \boldsymbol{v})_i = \varepsilon_{ijk}\omega_j v_k. \tag{6.8}$$

For the extension to four-dimensional spacetime, it is helpful to think of the rotation as occurring in the plane perpendicular to the angular velocity vector $\boldsymbol{\omega}$. Thus rewrite (6.8) as

$$dv_i/dt = -\Omega_{ik}v_k, \tag{6.9}$$

where

$$\Omega_{jk} = -\Omega_{kj} = \omega_i\varepsilon_{ijk} \tag{6.10}$$

has non-zero components only in the plane of the rotation. In other words, to speak of "a rotation in the (1, 2)-plane" is more useful than to speak of a rotation about the 3-axis. The concept of "plane of rotation" carries over to four dimensions. There a rotation in the (1, 2)-plane will leave constant not only the v_3 but also the v_0 component of the velocity. The four-dimensional definition of a rotation is

Mathematics of rotation in spacetime

$$\frac{dv^\mu}{d\tau} = -\Omega^{\mu\nu}v_\nu, \qquad \text{with} \qquad \Omega^{\mu\nu} = -\Omega^{\nu\mu}. \tag{6.11}$$

To test the appropriateness of this definition of a generalized rotation or infinitesimal Lorentz transformation, verify that it leaves invariant the length of the 4-vector:

$$d(v_\mu v^\mu)/d\tau = 2v_\mu(dv^\mu/d\tau) = -2\Omega^{\mu\nu}v_\mu v_\nu = 0. \tag{6.12}$$

The last expression vanishes because $\Omega^{\mu\nu}$ is antisymmetric, whereas $v_\mu v_\nu$ is symmetric. Note also that the antisymmetric tensor $\Omega^{\mu\nu}$ ("rotation matrix"; "infinitesimal Lorentz transformation") has $4 \times 3/2 = 6$ independent components. This number agrees with the number of components in a finite Lorentz transformation (three parameters for rotations, plus three parameters for the components of a boost). The "infinitesimal Lorentz transformation" here must (1) generate the appropriate Lorentz transformation in the timelike plane spanned by the 4-velocity and the 4-acceleration, and (2) exclude a rotation in any other plane, in particular, in any spacelike plane. The unique answer to these requirements is

$$\Omega^{\mu\nu} = a^\mu u^\nu - a^\nu u^\mu; \qquad \text{i.e., } \boldsymbol{\Omega} = \boldsymbol{a} \wedge \boldsymbol{u}. \tag{6.13}$$

Apply this rotation to a spacelike vector $\boldsymbol{w}$ orthogonal to $\boldsymbol{u}$ and $\boldsymbol{a}$, ($\boldsymbol{u} \cdot \boldsymbol{w} = 0$ and $\boldsymbol{a} \cdot \boldsymbol{w} = 0$). Immediately compute $\Omega^{\mu\nu}w_\nu = 0$. Thus verify the absence of any space rotation. Now check the over-all normalization of $\Omega^{\mu\nu}$ in equation (6.13). Apply the infinitesimal Lorentz transformation to the velocity 4-vector $\boldsymbol{u}$ of the observer. Thus insert $v^\mu = u^\mu$ in (6.11). It then reads

$$du^\mu/d\tau \equiv a^\mu = u^\mu(a^\nu u_\nu) - a^\mu(u^\nu u_\nu) = a^\mu.$$

This result is an identity, since $\boldsymbol{u} \cdot \boldsymbol{u} = -1$ and $\boldsymbol{u} \cdot \boldsymbol{a} = 0$.

A vector $\boldsymbol{v}$ that undergoes the indicated infinitesimal Lorentz transformation,

Fermi-Walker law of transport for "nonrotating" tetrad of basis vectors carried by an accelerated observer

$$dv^\mu/d\tau = (u^\mu a^\nu - u^\nu a^\mu)v_\nu, \tag{6.14}$$

is said to experience "Fermi-Walker transport" along the world line of the observer.

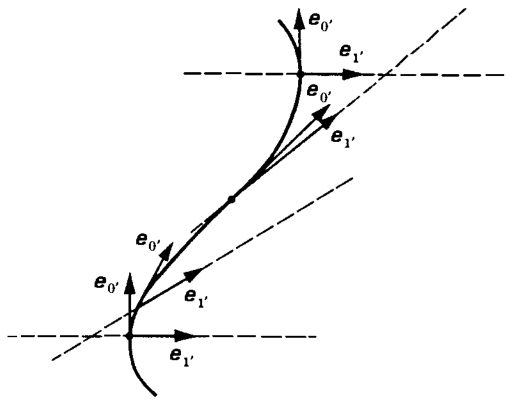

Figure 6.3.
Construction of spacelike hyperplanes (dashed) orthogonal to the world line (heavy line) of an accelerated particle at selected moments along that world line. Note crossing of hyperplanes at distance $g^{-1}(\tau)$ (time-dependent acceleration!) from the world line.

The natural moving frame associated with an accelerated observer consists of four orthonormal vectors, each of which is Fermi-Walker transported along the world line and one of which is $\boldsymbol{e}_{0'} = \boldsymbol{u}$ (the 4-velocity of the observer). Fermi-Walker transport of the space basis vectors $\boldsymbol{e}_{j'}$ can be achieved in practice by attaching them to gyroscopes (see Box 6.2 and exercise 6.9).

§6.6. THE LOCAL COORDINATE SYSTEM OF AN ACCELERATED OBSERVER

Tetrad used to construct "local coordinate system of accelerated observer"

Extend this moving frame or "infinitesimal coordinate system" to a "local coordinate system" covering a finite domain. Such local coordinates can escape none of the problems encountered in "hyperbolic motion" (Figure 6.1) and "briefly accelerated motion" (Figure 6.2). Therefore the local coordinate system has to be restricted to a region within a distance g^{-1} of the observer, where these problems do not arise. Figure 6.3 illustrates the construction of the local coordinates $\xi^{\mu'}$. At any given proper time τ the observer sits at a specific event $\mathscr{P}(\tau)$ along his world line. Let the displacement vector, from the origin of the original inertial frame to his position $\mathscr{P}(\tau)$, be $\boldsymbol{z}(\tau)$. At $\mathscr{P}(\tau)$ the observer has three spacelike basis vectors $\boldsymbol{e}_{1'}(\tau)$, $\boldsymbol{e}_{2'}(\tau)$, $\boldsymbol{e}_{3'}(\tau)$. The point $\mathscr{P}(\tau)$ plus those basis vectors define a spacelike hyperplane. The typical point of this hyperplane can be represented in the form

$$\begin{aligned} \boldsymbol{x} &= \xi^{1'}\boldsymbol{e}_{1'}(\tau) + \xi^{2'}\boldsymbol{e}_{2'}(\tau) + \xi^{3'}\boldsymbol{e}_{3'}(\tau) + \boldsymbol{z}(\tau) \\ &= \text{(separation vector from origin of original inertial frame).} \end{aligned} \tag{6.15}$$

Here the three numbers $\xi^{k'}$ play the role of Euclidean coordinates in the hyperplane. This hyperplane advances as proper time unrolls. Eventually the hyperplane cuts through the event $\mathscr{P}_0$ to which it is desired to assign coordinates. *Assign to this event as coordinates the numbers* $\xi^{0'} = \tau$, $\xi^{k'}$ *given by (6.15). Call these four numbers* *"coordinates relative to the accelerated observer."* In detail, the prescription for the determination of these four coordinates consists of the four equations

$$x^\mu = \xi^{k'}[e_{k'}(\tau)]^\mu + z^\mu(\tau), \tag{6.16}$$

in which the x^μ are considered as known, and the coordinates τ, $\xi^{k'}$ are considered unknowns.

At a certain distance from the accelerated world line, successive spacelike hyperplanes, instead of advancing with increasing τ, will be retrogressing. At this distance, and at greater distances, the concept of "coordinates relative to the accelerated observer" becomes ambiguous and has to be abandoned. To evaluate this distance, note that any sufficiently short section of the world line can be approximated by a hyperbola ("hyperbolic motion with acceleration g"), where the time-dependent acceleration $g(\tau)$ is given by the equation $g^2 = a^\mu a_\mu$.

Local coordinate system for uniformly accelerated observer

Apply the above general prescription to hyperbolic motion, arriving at the equations

$$\begin{aligned} x^0 &= (g^{-1} + \xi^{1'})\sinh(g\xi^{0'}), \\ x^1 &= (g^{-1} + \xi^{1'})\cosh(g\xi^{0'}), \\ x^2 &= \xi^{2'}, \\ x^3 &= \xi^{3'}. \end{aligned} \tag{6.17}$$

The surfaces of constant $\xi^{0'}$ are the hyperplanes with $x^0/x^1 = \tanh g\xi^{0'}$ sketched in Figure 6.4. Substitute expressions (6.17) into the Minkowski formula for the line element to find

$$\begin{aligned} ds^2 &= \eta_{\mu\nu}\, dx^\mu\, dx^\nu \\ &= -(1 + g\xi^{1'})^2(d\xi^{0'})^2 + (d\xi^{1'})^2 + (d\xi^{2'})^2 + (d\xi^{3'})^2. \end{aligned} \tag{6.18}$$

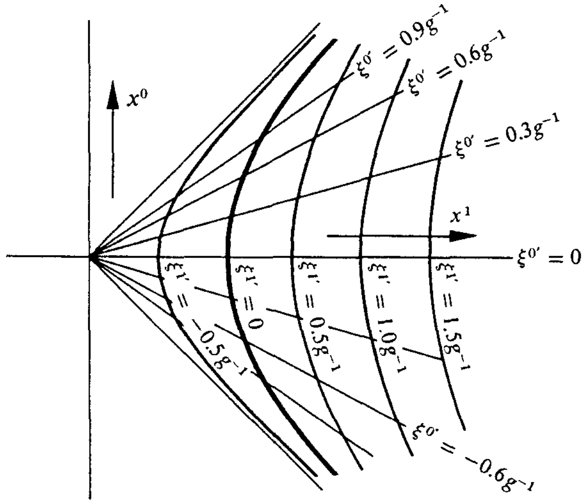

Figure 6.4.
Local coordinate system associated with an observer in hyperbolic motion (heavy black world line). The local coordinate system fails for $\xi^{1'}$ less than $-g^{-1}$.

The coefficients of $d\xi^{\mu'}\, d\xi^{\nu'}$ in this expansion are not the standard Lorentz metric components. The reason is clear. The $\xi^{\mu'}$ do not form an inertial coordinate system. However, at the position of the observer, $\xi^{1'} = 0$, the coefficients reduce to the standard form. Therefore these "local coordinates" approximate a Lorentz coordinate system in the immediate neighborhood of the observer.

EXERCISES

Exercise 6.6. CLOCK RATES VERSUS COORDINATE TIME IN ACCELERATED COORDINATES

Let a clock be attached to each grid point, $(\xi^{1'}, \xi^{2'}, \xi^{3'})$ = constant, of the local coordinate system of an accelerated observer. Assume for simplicity that the observer is in hyperbolic motion. Use equation (6.18) to show that proper time as measured by a lattice clock differs from coordinate time at its lattice point:

$$d\tau/d\xi^{0'} = 1 + g\xi^{1'}.$$

(Of course, very near the observer, at $\xi^{1'} \ll g^{-1}$, the discrepancy is negligible.)

Exercise 6.7. ACCELERATION OF LATTICE POINTS IN ACCELERATED COORDINATES

Let an accelerometer be attached to each grid point of the local coordinates of an observer in hyperbolic motion. Calculate the magnitude of the acceleration measured by the accelerometer at $(\xi^{1'}, \xi^{2'}, \xi^{3'})$.

Exercise 6.8. OBSERVER WITH ROTATING TETRAD

An observer moving along an arbitrarily accelerated world line chooses *not* to Fermi-Walker transport his orthonormal tetrad. Instead, he allows it to rotate. The antisymmetric rotation tensor $\boldsymbol{\Omega}$ that enters into his transport law

$$d\boldsymbol{e}_{\alpha'}/d\tau = -\boldsymbol{\Omega}\cdot\boldsymbol{e}_{\alpha'} \tag{6.19}$$

splits into a Fermi-Walker part plus a spatial rotation part:

$$\Omega^{\mu\nu} = \underbrace{a^\mu u^\nu - a^\nu u^\mu}_{\Omega^{\mu\nu}_{(FW)}} + \underbrace{u_\alpha \omega_\beta \epsilon^{\alpha\beta\mu\nu}}_{\Omega^{\mu\nu}_{(SR)}} \tag{6.20}$$

$$\boldsymbol{\omega} = \text{a vector orthogonal to 4-velocity } \boldsymbol{u}.$$

(a) The observer chooses his time basis vector to be $\boldsymbol{e}_{0'} = \boldsymbol{u}$. Show that this choice is permitted by his transport law (6.19), (6.20).

(b) Show that $\Omega^{\mu\nu}_{(SR)}$ produces a rotation in the plane perpendicular to $\boldsymbol{u}$ and $\boldsymbol{\omega}$—i.e., that

$$\boldsymbol{\Omega}_{(SR)}\cdot\boldsymbol{u} = 0, \qquad \boldsymbol{\Omega}_{(SR)}\cdot\boldsymbol{\omega} = 0. \tag{6.21}$$

(c) Suppose the accelerated observer Fermi-Walker transports a second orthonormal tetrad $\boldsymbol{e}_{\alpha''}$. Show that the space vectors of his first tetrad rotate relative to those of his second tetrad with angular velocity vector equal to $\boldsymbol{\omega}$. *Hint:* At a moment when the tetrads coincide, show that (in three-dimensional notation, referring to the 3-space orthogonal to the observer's world line):

$$d(\boldsymbol{e}_{j'} - \boldsymbol{e}_{j''})/d\tau = \boldsymbol{\omega}\times\boldsymbol{e}_{j'}. \tag{6.22}$$

(d) The observer uses the same prescription [equation (6.16)] to set up local coordinates based on his rotating tetrad as for his Fermi-Walker tetrad. Pick an event $\mathscr{Q}$ on the observer's world line, set $\tau = 0$ there, and choose the original inertial frame of prescription (6.16) so (1) it comoves with the accelerated observer at $\mathscr{Q}$, (2) its origin is at $\mathscr{Q}$, and (3) its axes coincide with the accelerated axes at $\mathscr{Q}$. Show that these conditions translate into

$$z^\mu(0) = 0, \qquad \boldsymbol{e}_{\alpha'}(0) = \boldsymbol{e}_\alpha. \tag{6.23}$$

(e) Show that near $\mathscr{Q}$, equations (6.16) for the rotating, accelerated coordinates reduce to:

$$\begin{aligned} x^0 &= \xi^{0'} + a_{k'}\xi^{k'}\xi^{0'} + O([\xi^{\alpha'}]^3); \\ x^j &= \xi^{j'} + \frac{1}{2}a^j\xi^{0'2} + \epsilon^{jkl}\omega^k\xi^{l'}\xi^{0'} + O([\xi^{\alpha'}]^3). \end{aligned} \tag{6.24}$$

(f) A freely moving particle passes through the event $\mathscr{Q}$ with ordinary velocity $\boldsymbol{v}$ as measured in the inertial frame. By transforming its straight world line $x^j = v^j x^0$ to the accelerated, rotating coordinates, show that its coordinate velocity and acceleration there are:

$$\begin{aligned} (d\xi^{j'}/d\xi^{0'})_{\text{at } \mathscr{Q}} &= v^j; \\ (d^2\xi^{j'}/d\xi^{0'2})_{\text{at } \mathscr{Q}} &= \underbrace{-a^j}_{\substack{\text{inertial}\\ \text{acceleration}}} - \underbrace{2\epsilon^{jkl}\omega^k v^l}_{\substack{\text{Coriolis}\\ \text{acceleration}}} + \underbrace{2v^j a^k v^k}_{\substack{\text{relativisitc}\\ \text{correction to}\\ \text{inertial acceleration}}}. \end{aligned} \tag{6.25}$$

Exercise 6.9. THOMAS PRECESSION

Consider a spinning body (gyroscope, electron, . . .) that accelerates because forces act at its center of mass. Such forces produce no torque; so they leave the body's intrinsic angular-momentum vector $\boldsymbol{S}$ unchanged, except for the unique rotation in the $\boldsymbol{u} \wedge \boldsymbol{a}$ plane required to keep $\boldsymbol{S}$ orthogonal to the 4-velocity $\boldsymbol{u}$. Mathematically speaking, the body Fermi-Walker transports its angular momentum (no rotation in planes other than $\boldsymbol{u} \wedge \boldsymbol{a}$):

$$d\boldsymbol{S}/d\tau = (\boldsymbol{u} \wedge \boldsymbol{a}) \cdot \boldsymbol{S}. \tag{6.26}$$

This transport law applies to a spinning electron that moves in a circular orbit of radius r around an atomic nucleus. As seen in the laboratory frame, the electron moves in the x, y-plane with constant angular velocity, ω. At time $t = 0$, the electron is at $x = r$, $y = 0$; and its spin (as treated classically) has components

$$S^0 = 0, \qquad S^x = \frac{1}{\sqrt{2}}\hbar, \qquad S^y = 0, \qquad S^z = \frac{1}{2}\hbar.$$

Calculate the subsequent behavior of the spin as a function of laboratory time, $S^\mu(t)$. Answer:

$$\begin{aligned} S^x &= \frac{1}{\sqrt{2}}\hbar\,(\cos\omega t\cos\omega\gamma t + \gamma\sin\omega t\sin\omega\gamma t); \\ S^y &= \frac{1}{\sqrt{2}}\hbar\,(\sin\omega t\cos\omega\gamma t - \gamma\cos\omega t\sin\omega\gamma t); \\ S^z &= \frac{1}{2}\hbar; \qquad S^0 = -\frac{1}{\sqrt{2}}\hbar\, v\gamma\sin\omega\gamma t; \\ v &= \omega r; \qquad \gamma = (1 - v^2)^{-1/2}. \end{aligned} \tag{6.27}$$

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Rewrite the time-dependent spatial part of this as

$$S^x + iS^y = \frac{\hbar}{\sqrt{2}}[e^{-i(\gamma-1)\omega t} + i(1-\gamma)\sin(\omega\gamma t)e^{i\omega t}]. \tag{6.28}$$

The first term rotates steadily in a retrograde direction with angular velocity

$$\begin{aligned}\omega_{\text{Thomas}} &= (\gamma - 1)\omega \\ &\approx \frac{1}{2}v^2\omega \ll \omega \text{ if } v \ll 1.\end{aligned} \tag{6.29}$$

It is called the Thomas precession. The second term rotates in a righthanded manner for part of an orbit ($0 < \omega\gamma t < \pi$) and in a lefthanded manner for the rest ($\pi < \omega\gamma t < 2\pi$). Averaged in time, it does nothing. Moreover, in an atom it is very small ($\gamma - 1 \ll 1$). It must be present, superimposed on the Thomas precession, in order to keep

$$\mathbf{S}\cdot\mathbf{u} = \boldsymbol{S}\cdot\boldsymbol{u} - S^0u^0 = 0, \tag{6.30}$$

and

$$\mathbf{S}^2 = \boldsymbol{S}^2 - (S^0)^2 = 3\hbar^2/4 = \text{constant}. \tag{6.31}$$

It comes into play with righthanded rotation when $\boldsymbol{S}\cdot\boldsymbol{u}$ is negative; it goes out of play when $\boldsymbol{S}\cdot\boldsymbol{u} = 0$; and it returns with lefthanded rotation when $\boldsymbol{S}\cdot\boldsymbol{u}$ turns positive.

The Thomas precession can be understood, alternatively, as a spatial rotation that results from the combination of successive boosts in slightly different directions. [See, e.g., exercise 103 of Taylor and Wheeler (1966).] For an alternative derivation of the Thomas precession (6.29) from "spinor formalism," see §41.4.